

Laplace for Second-Order Denominators and Intro to TFs

EGR 345

GVSU

Outline

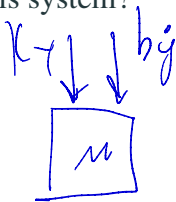
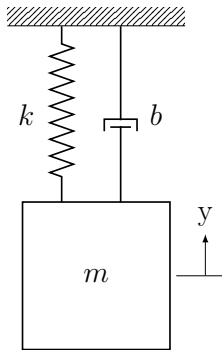
- quick review
- Laplace for second-order denominators
- Two things we need for lab:
 - transfer functions
 - plants

Mass/Spring/Damper Response

- how does this system respond to an initial displacement?
 - by that I mean, find $y(t)$
- what is the differential equation associated with this system?

→ decaying
sine wave

$$N^2$$
$$m\ddot{y} = -ky - b\dot{y}$$



$$m\ddot{y} + b\dot{y} + ky = 0$$

What is ζ ?

$$\frac{b_{cr}^2}{m^2} - 4\frac{k}{m} = 0$$

From Quadratic
Formula

$$ms^2 + bs + k = (s^2 + 2\zeta\omega_n s + \omega_n^2)/m$$

$$s^2 + \frac{b}{m}s + \frac{k}{m} = s^2 + 2\zeta\omega_n s + \omega_n^2$$

$$\frac{b}{m} = ?$$

$$\frac{k}{m} = ? \quad \omega_n^2$$

$$\zeta = \frac{b}{b_{cr}}$$
$$b_{cr} = 2\sqrt{km}$$

$b > b_{cr} \rightarrow$ distinct
($\zeta > 1$) real
roots

$b < b_{cr} \rightarrow$ complex
($\zeta < 1$) roots

$$\xi = \frac{b}{b_{cr}} = \frac{b}{2\sqrt{km}}$$

$$2\xi\omega_n \stackrel{?}{=} \frac{b}{m}$$

~~$$2 \frac{b}{2\sqrt{km}} \cdot \sqrt{\frac{k}{m}} \stackrel{?}{=} \frac{b}{m}$$~~

$$\frac{b}{\sqrt{m} \cdot \sqrt{a}} = \frac{b}{m} \quad \checkmark$$

Finishing Up Second-Order Laplace Denominators

$$Y(s) = \frac{y_0(s + b/m)}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

Laplace with complex 2nd order roots

Key Idea: if we have complex roots, we must use rows 8 **and** 9

$$Y(s) = \frac{y_0(s + b/m)}{s^2 + 2\zeta\omega_n s + \omega_n^2} - \frac{C_1\omega}{(s+a)^2 + \omega^2} + \frac{C_2(s+a)}{(s+a)^2 + \omega^2}$$

Known: $m, b, k \rightarrow \zeta, \omega_n$

Unknown: C_1, C_2, a , and ω

from the denominator:

$$s^2 + 2\zeta\omega_n s + \omega_n^2 = (s+a)^2 + \omega^2$$

$$\underline{s^2 + 2\zeta\omega_n s + \omega_n^2} = \underline{s^2 + 2as + a^2 + \omega^2}$$

$$s^0: \omega_n^2 = a^2 + \omega^2$$

$$s^1: 2\zeta\omega_n = 2a \rightarrow a = \zeta\omega_n$$

$$s^2: 1 = 1 \quad \checkmark$$

(next slide)

$$\omega_n^2 = q^2 + \omega^2$$

$$\omega_n^2 = \gamma^2 \omega_n^2 + \omega^2$$

$$\omega_n^2 - \gamma^2 \omega_n^2 = \omega^2$$

$$\omega^2 = \omega_n^2 (1 - \gamma^2)$$

$$\omega \equiv \omega_d = \omega_n \sqrt{1 - \gamma^2}$$

$$a = \gamma \omega_n$$

$$\gamma^2 + 2\gamma \omega_n s + \omega_n^2 = s^2 + 2as + a^2 + \omega^2$$

$$Y(s) = \frac{y_0(s + b/n)}{s^2 + 2\zeta\omega_n s + \omega_n^2} = \frac{C_1 \omega + C_2(s+a)}{(s+a)^2 + \omega^2}$$

we now know a and ω ; still need C_1 and C_2

$$y_0(s + b/n) = C_1 \omega + C_2(s+a)$$

$$\underline{y_0 s} + \underline{y_0 \left(\frac{b}{n}\right)} = \underline{C_1 \omega} + \underline{C_2 s} + \underline{C_2 a}$$

$$\underline{y_0 s} + \underline{y_0 \left(\frac{b}{n} \right)} = \underline{C_1 \omega} + \underline{C_2 s} + \underline{C_2 a}$$

$$C_2 = y_0$$

$$s^0: y_0 \left(\frac{b}{n} \right) = C_1 \omega + C_2 a$$

$$C_1 = \frac{y_0}{\omega} \left(\frac{b}{n} - a \right)$$

$$s^1: \cancel{y_0 s} = \cancel{C_2 s} \rightarrow C_2 = y_0$$

$$y_0 \frac{b}{n} = C_1 \omega + \cancel{y_0 a}$$

$$C_1 = \frac{1}{\omega} \left(y_0 \frac{b}{n} - y_0 a \right)$$

From rows 8 & 9 of the Laplace table:

$$y(t) = C_1 e^{-at} \sin(\omega t) + C_2 e^{-at} \cos(\omega t)$$

$$a = \zeta \omega_n$$

$$y(0) = y_0$$

$$\omega = \omega_d = \omega_n \sqrt{1 - \zeta^2}$$

$$C_1 = \frac{y_0}{\omega} \left(\frac{b}{n} - a \right)$$

$$C_2 = y_0$$

Naived Verification

just

→ see note book

Transfer Functions and Plants

What is a transfer function?

- my software (which we will start using in lab) is designed around transfer functions and plants

a transfer function is an input/output model in the Laplace domain that comes from the differential equation

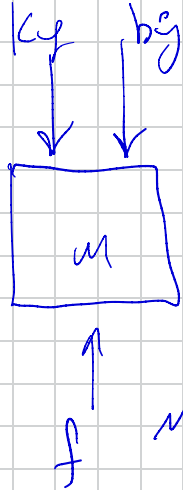
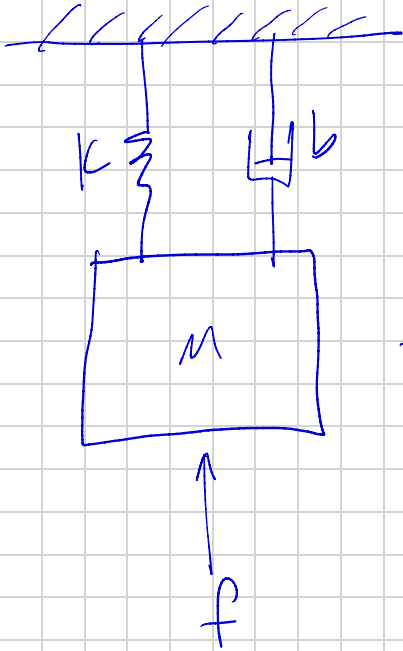
$$G(s) = \frac{Out(s)}{In(s)}$$



$G(s)$ is the
transfer
function

$$G(s) = \frac{Out(s)}{I_n(s)}$$

$$Out(s) = G(s) \cdot I_n(s)$$



$$m\ddot{y} = -k_y - b_y + f$$

$$m\ddot{y} + b_y + k_y = f$$

$$\mathcal{L}(m\ddot{y} + b\dot{y} + ky = f)$$

assuming zero initial conditions
 $y(0) = 0; \dot{y}(0) = 0$

$$ms^2 Y(s) + bs Y(s) + k Y(s) = F(s)$$

$$Y(s)(ms^2 + bs + k) = F(s)$$

$$\frac{\text{Out}(s)}{\text{In}(s)} = G(s) = \frac{Y(s)}{F(s)} = \frac{1}{ms^2 + bs + k} = \frac{1/m}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

Transfer Function for MSD sys. based on earlier slides (2-12)

What is a plant?

in feedback controls, plant refers to the physical system you are analyzing or designing a control system for

→ a plant is the physical system your TF represents

OT Idea

— just like a TF has an input and an output, a plant has an actuator and a sensor

step input

